s10_vol_rank_tilt

equity strategy | family: equity-volatility | status: research | derived from `s9b_inverse_vol`

What this strategy does

Strategy 6 weights x (1.02 - prior-quarter ivol percentile)^2 inside each neutralization cell

Why it should work. PEAD alpha per dollar is highest in calm stocks; rank-weight toward them without depending on vol levels.

Design notes (from the strategy's source file).

Strategy 10 -- Strategy 6 tilted toward calm stocks by volatility RANK (research).

Strategy 9b sized each event by 1/idiosyncratic-vol (Sharpe 1.30 -> 1.47). The attribution work showed that PEAD alpha per dollar is concentrated in low-volatility names and almost absent in the wildest tercile, and that the 1/vol weight, being linear in the vol LEVEL, only mildly demotes the high-vol tail. Weighting by (1.02 - vol percentile)^2 -- where the percentile is measured against the previous quarter's cross-section within the stock's size quintile -- demotes it hard while staying smooth:

Sharpe 1.30 (S6) -> 1.47 (9b) -> 1.81; max drawdown -16.6% -> -11.6%; every era improves split-sample check: exponent picked on 1996-2004 data alone (best k ~ 3) lifts 2005-2013 Sharpe 0.93 -> 1.10, the same as the full-sample best, so the gain is not an artifact of one period.

Lower volatility buys room: `s10_vol_rank_tilt_levered` spends it on 1.7x the base unit and a 4x gross cap for ~9% a year at ~5% vol (a market-neutral book clearing the S&P 500's return in this sample). More sizing than that stops paying: the 5%-of-ADV liquidity cap starts to bind and 2008-13 Sharpe decays.

STATUS research: the exponent and sizing were chosen after seeing 9b's full-sample results (the split-sample check above is the only out-of-sample evidence), and the levered variant leans on small-cap capacity.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
beta_overlay	0.0

Performance

metric	s10 vol rank tilt	market (CRSP VW)
Annualized return	7.12%	9.94%
Annualized volatility	3.83%	19.35%
Sharpe (NAV-pct)	1.81	0.59
Sortino	2.77	
Max drawdown	-11.6%	-55.5%

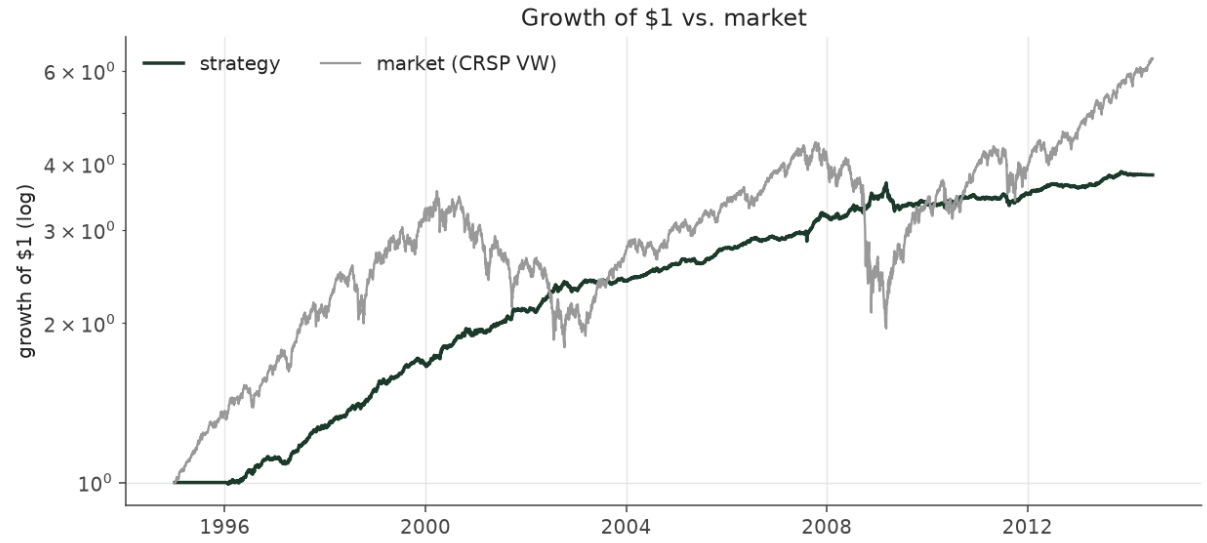
Calmar	0.61	
Final NAV ($10M start)	$38,141,081	
Worst day	-2.55%	
% positive days	52.7%	
Sharpe 1996-2001	2.94	
Sharpe 2002-2007	1.89	
Sharpe 2008-2013	0.78	

Correlation of daily returns with the market: **-0.01**.

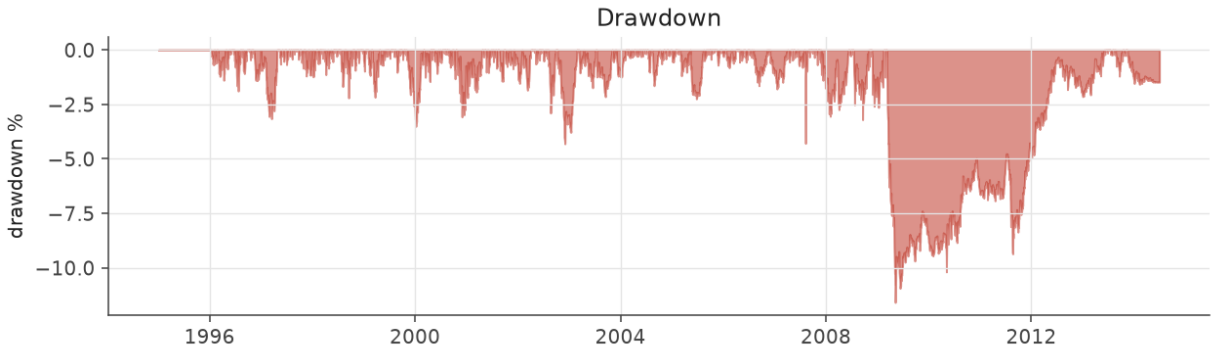
• best Sharpe among equity: this strategy; this one: 1.81

• best return among equity: `s6_beta_overlay_aggressive` (0.15); this one: 0.0712

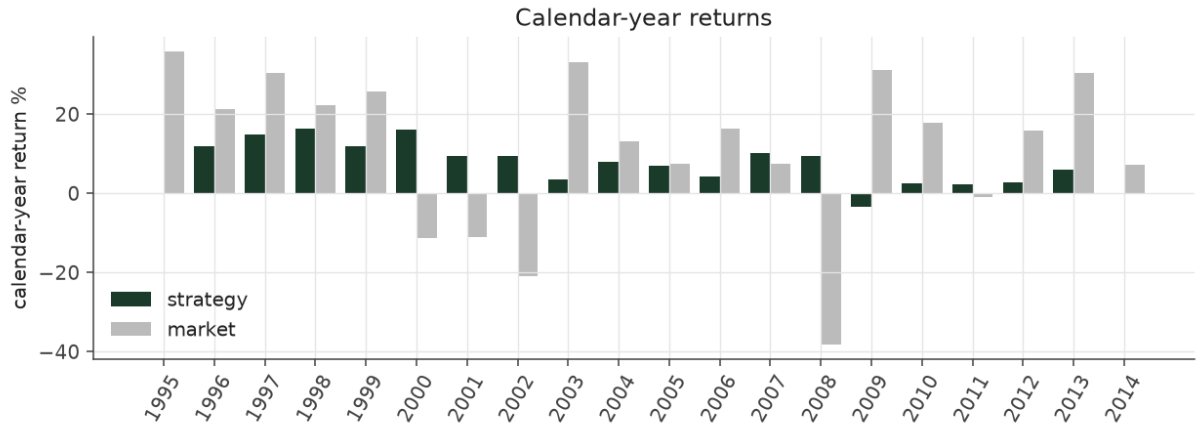
• best Calmar among equity: this strategy; this one: 0.614



NAV (log scale)



Drawdown from running peak

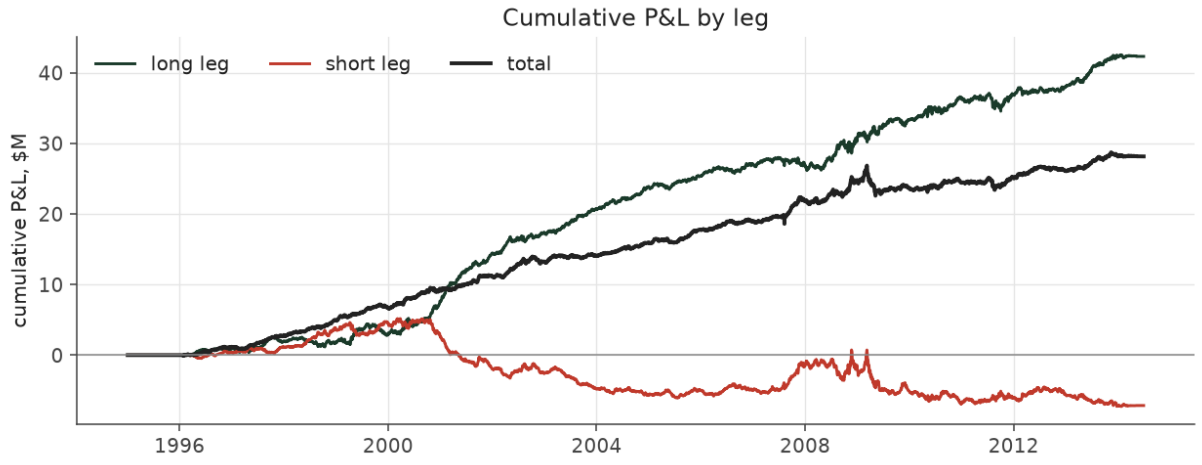


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	42.31	-7.14
P&L after costs ($M)	38.92	-10.78
of which common (average-event) component ($M)	27.61	-27.83

of which excess / PEAD component ($M)	14.70	20.69
excess minus costs ($M)	11.31	17.05
gross return per $ traded (bp)	244.60	-40.33
positions in the money (hit rate)	0.56	0.49
stand-alone Sharpe (leg only)	1.48	-0.16

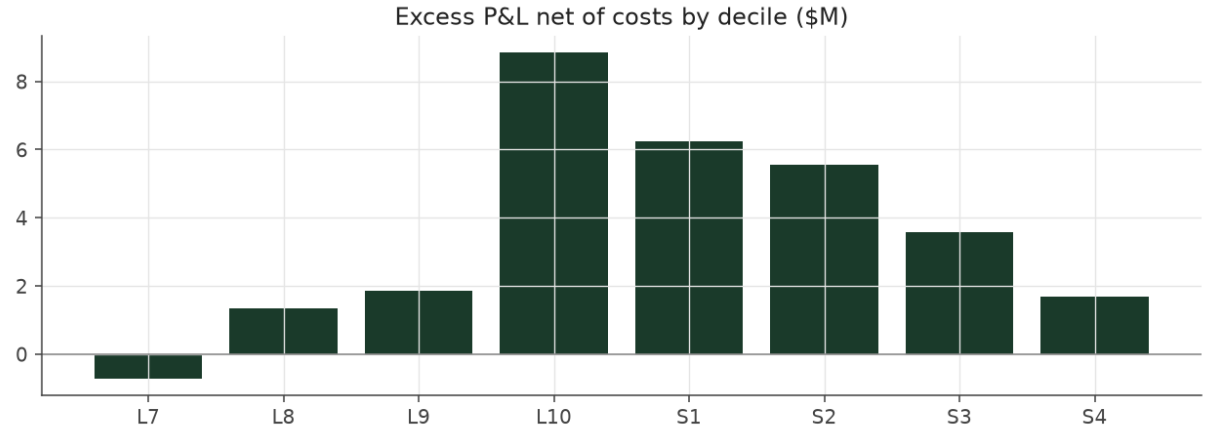
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-7.1M before costs) while the long leg earns +42.3M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 27.6M / 27.8M. Stripping that common component, the surprise signal itself earned +14.7M on the long side and +20.7M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.48 long / -0.16 short vs 1.81 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	8,029	158.66	2.31	-0.42	0.31	-0.73	-2.6%	-45.85
long	8	16,588	415.11	8.24	2.15	0.82	1.34	4.7%	32.24
long	9	17,747	535.73	9.62	2.89	1.03	1.85	6.5%	34.61
long	10	17,174	620.10	18.75	10.07	1.22	8.85	31.2%	142.70
short	1	17,169	611.53	-2.46	7.54	1.31	6.23	22.0%	101.86
short	2	17,281	543.16	-3.69	6.66	1.11	5.55	19.6%	102.21
short	3	15,970	433.58	-3.41	4.45	0.87	3.58	12.6%	82.48
short	4	9,171	182.37	-1.23	2.04	0.35	1.69	6.0%	92.85



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	14,213	258.09	5.07	6.67	1.01	5.66	20.0%	219.35
2	22,885	608.55	6.10	8.01	1.76	6.25	22.0%	102.75

3	25,501	801.21	7.23	8.75	1.75	7.00	24.7%	87.41
4	27,583	905.05	6.03	7.37	1.48	5.89	20.8%	65.06
5	28,947	927.33	3.71	4.59	1.03	3.56	12.5%	38.35

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,650	208.10	4.29	5.68	1.04	4.64	16.3%	222.79
2	23,084	716.85	6.64	9.78	2.15	7.63	26.9%	106.44
3	25,851	912.90	8.09	10.05	1.83	8.22	29.0%	90.08
4	27,665	861.29	5.41	5.79	1.21	4.59	16.2%	53.26
5	28,874	800.98	3.71	4.07	0.80	3.27	11.5%	40.82

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	45,855	1,393.64	2.33	9.87	2.49	7.38	26.0%	52.97
41-60d	25,652	807.71	6.04	5.82	1.65	4.17	14.7%	51.66
61-90d	35,324	966.11	8.08	9.29	2.05	7.24	25.5%	74.92
>90d	12,298	332.76	11.69	10.40	0.83	9.57	33.7%	287.61

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	33,895	660.14	11.04	12.55	1.31	11.24	39.6%	170.30
2002-07	39,956	1,201.49	10.83	12.96	2.37	10.60	37.4%	88.18
2008-13	45,278	1,638.60	6.27	9.88	3.35	6.53	23.0%	39.83

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	22,094	348.54	3.31	3.97	0.76	3.22	11.3%	92.28
Chems	2,590	232.45	1.79	2.14	0.40	1.74	6.1%	74.80
Durbl	2,843	247.37	1.98	2.43	0.44	1.98	7.0%	80.17
Enrgy	6,194	273.83	3.43	4.04	0.51	3.52	12.4%	128.69
Hlth	11,696	332.40	2.17	2.80	0.69	2.11	7.4%	63.45
Manuf	12,034	318.17	2.01	2.84	0.66	2.18	7.7%	68.42
Money	17,583	315.04	2.93	3.72	0.71	3.01	10.6%	95.42
NoDur	5,081	282.00	1.73	2.19	0.53	1.66	5.9%	59.01
Other	18,377	338.29	3.08	3.92	0.76	3.15	11.1%	93.22
Shops	11,984	326.93	3.01	3.63	0.68	2.96	10.4%	90.41

Telcm	3,981	259.18	1.94	2.38	0.46	1.92	6.8%	74.07
Utils	4,672	226.03	0.76	1.33	0.41	0.92	3.2%	40.51

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	39,523	293.11	3.29	4.05	0.55	3.50	12.4%	119.56
low vol	40,091	1,985.36	14.71	19.31	4.09	15.22	53.6%	76.64
mid vol	37,903	1,165.50	9.67	11.45	2.26	9.18	32.4%	78.80

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	7.12%	0.00%
0.5%	1.8	6.85%	0.26%
1.0%	3.5	6.58%	0.53%
2.0%	7.1	5.99%	1.12%
5.0%	17.7	3.74%	3.37%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.